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[24ESEC12]

FIRST SEMESTER B. TECH EXAMINATION, JANUARY-2025

ELECTRICAL AND ELECTRONICS SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	RBT	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	What is duplexer in RADAR system?	L1	CO1	**02**
	b.	What Is Skin Effect?	L1	CO3	**02**
	c.	What are the disadvantage of satellite communication	L1	CO2	**02**
	d.	What is meant by attitude control w.r.t satellite system.	L1	CO2	**02**
	e.	Define hand off w.r.t wireless communications.	L1	CO2	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain amplitude modulation and what are its advantages?	L2	CO1	**05**
		OR			
	b.	List the different sensors and Their functions in the mobile phone handset.	L2	CO1	**05**
3.	**a.**	Explain AM TRF Receiver with block diagram?	L2	CO1	**05**
		OR			
	b.	What are the function of RF amplifier and mixer in receiver?	L2	CO1	**05**
4.	**a.**	Explain briefly three parameters taken care while designing a smart watch.	L2	CO2	**05**
		OR			
	b.	Explain briefly the structure of IoT.	L2	CO2	**05**
5.	**a.**	compare HVAC and HVDC transmission.	L2	CO3	**05**
		OR			
	b.	What are the environmental impact and drawbacks of hydraulic power plants?	L2	CO3	**05**
6.	**a.**	What is power factor? what are the significance of power factor.	L2	CO4	**05**
		OR			
	b.	Briefly explain different power losses.	L2	CO4	**05**

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7.	**a.**	What are the main function of BTS and BSC in GSM network.	L2	CO2	**05**
		OR			
	b.	List any five comparisons between conventional grid and Smart grid.	L2	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Explain the AM transmitter with block diagram and their advantages.	L2	CO1	**10**
		OR			
	b.	Explain frequency modulation transmitter and List FM advantages and disadvantages?	L2	CO1	**10**
9.	**a.**	List three segments of satellite communication explain them briefly.	L2	CO2	**10**
		OR			
	b.	List three subsystems of GSM architecture and briefly explain.	L2	CO2	**10**
10.	**a.**	Explain technological challenges of IoT.	L2	CO2	**10**
		OR			
	b.	List the criticisms and controversies of IoT.	L2	CO2	**10**
11.	**a.**	Explain the operation of smart watch using its block diagram.	L2	CO2	**10**
		OR			
	b.	Explain different methods of power savings.	L2	CO4	**10**
12.	**a.**	What is smart grid, explain briefly and list its advantages.	L2	CO4	**10**
		OR			
	b.	Explain the working of transformer and list its applications.	L2	CO4	**10**
13.	**a.**	(i)List the types of power plants. (ii)what are the main disadvantages of nuclear power plants	L2	CO3	**10**
		OR			
	b.	Explain thermal power plant using its schematic diagram.	L2	CO3	**10**